SECOND PROJECT PROGRESS REPORT

RetailLens

Retail Sales Forecasting & Analytics

Date: 29 September 2026 **Stage:** Evaluation and dashboard integration

Dataset: Store Item Demand Forecasting Challenge

1. Project Overview

RetailLens analyzes historical retail sales, compares forecasting models, and presents demand insights through a Streamlit dashboard to support inventory planning.

2. Progress So Far

- Created a data-cleaning pipeline and processed sales dataset.
- Implemented calendar, lag, and rolling features for sales forecasting.
- Built Prophet, SARIMA, and XGBoost model workflows and saved evaluation outputs.
- Developed a Streamlit dashboard with sales filters, charts, forecast/model views, downloads, and demand-based attention alerts.
- Added forecasting and machine-learning test suites; a passing test run has not yet been verified for this report.

3. Current Results

The saved detailed comparison covers Store 1 / Item 1 on a 90-day holdout. It is a single-series result, not a final ranking across the complete dataset.

Model	MAE	RMSE	MAPE	Coverage
Prophet (Default)	3.718	4.500	22.82%	97.78%
Prophet (Tuned)	3.709	4.476	21.87%	94.44%
SARIMA	4.494	5.386	29.23%	98.89%
XGBoost	4.921	6.074	10.47%	0.00%

The detailed report marks XGBoost as best, although tuned Prophet has lower MAE and RMSE in the same results. A separate ML report also gives different Prophet scores. Therefore, the final best model is not yet confirmed. XGBoost interval coverage is not available.

4. Next Work

1. Re-run all models on the same holdout dates and metric definitions; correct the inconsistent comparison and best-model selection.
2. Add prediction intervals for XGBoost and evaluate their coverage and width.
3. Test 30-, 60-, and 90-day forecasts across more store-item series, then summarize overall performance.
4. Run the test suites and verify the dashboard workflow, forecast selection, and alert labels.

5. Connect actual inventory data before presenting demand-based alerts as low-stock warnings.

5. Conclusion

The core forecasting modules and dashboard prototype are in place. The next phase will focus on consistent model evaluation, uncertainty estimates, broader testing, and verified dashboard behavior before selecting a final model.